Lesson 7:
Linux Users

Introduction

- Every user interacting with a Linux system has an account.
- But apart from standard users, there are a few other types of user accounts on Linux.
- User accounts are a key concept on Linux.
- Primarily, they help you manage access control on your system.
- In addition, they allow multiple users to log in at any given time, and they are critical for the security of a system.
- **Without user accounts, it would be hard to control who has access to what part of a Linux system.**
- **Linux has four main types of user accounts: Superuser, Regular, System and Service.**

User Account Identifiers

- A Linux system can have multiple user accounts, therefore, to be able to uniquely identify each account, Linux uses User Identifiers (UIDs) and Group Identifiers (GIDs).
- The very basis of any user account revolves around its UID.
- Furthermore, each user account is associated with a specific group and is therefore assigned a GID. A user account can belong to several groups.
- In addition to having user and group identifiers, every user account has its UID within a specific range that corresponds to the type of account it belongs to.

1. Superuser Account

- On Linux, the superuser account is popularly known as the **root**, the default administrator of your system.
- Each Linux system must at a *minimum have one root* user account.
- As part of the administrative privileges, the superuser account has <u>unlimited access and control of a system, including other users</u>.
- This is why you must always set a *strong password* for the root account.
- The superuser account always has the **UID 0** and is part of the **group ID 0** too.
- Unlike other accounts, the superuser has its home directory at the top-level directory, located at **/root.**

2. Regular Accounts

- All other users that are later added to a Linux system are known as regular accounts or standard accounts
- These types of accounts have limited access and control over the system but can gain administrative rights by using the **sudo** or **su** command.
- On modern Linux systems, regular users have UIDs with a four-digit number starting from 1000. In fact, the first user account you create on your system will usually have the UID 1000.
- By default, regular user accounts have their home directory within the **/home** directory.
- If you use the **ls** command to do a listing of the home directory, you'll get a good picture of the current users on your system.
- Another important attribute of regular accounts is that they have a default shell and on most Linux distros, this is the Bourne Again Shell (Bash).
- You can change your default shell if you prefer, for example, set it to the Z shell or C shell.

3. System Accounts

- The operating system creates system accounts during its installation.
- System accounts are <u>used for running operating system components</u> but **do not** run with superuser privileges.
- Most system accounts have their user identifiers in the range of 1 to 99, as a rule of thumb, two digits. In some cases, system accounts have three-digit UIDs between the numbers 500 and 999.
- An example of a system account is the Apache web server, which has the UID and GID number 48.
- System accounts don't have a dedicated home directory nor do they have a shell, because there is never a need for login in their lifetime.

4. Service Accounts

- Service accounts are a bit similar to system accounts: they are created whenever a service is installed onto your system.
- For example, when you install a SQL server database, the system creates a corresponding account for managing activities related to the SQL server.
- Just like system accounts, service accounts don't have a home folder or a default shell. They are assigned **/sbin/nologin** to refuse logins.
- Service accounts usually have a three-digit UID, between 100 and 999.
- In some cases, service accounts may have a four-digit UID just like regular accounts.
- Some prominent services that you will find on your system include systemd, GNOME Display Manager (GDM), CUPS printing service, etc.

Creating a user

Creating a user with a default setting:

- A user can be added by running the *useradd* command at the command prompt.
- After creating the user, set a password using the **passwd** utility, as follows:

```
[root@localhost bhargab]# useradd anirban
[root@localhost bhargab]# passwd anirban
Changing password for user anirban.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
```

- The system automatically assigns a UID, creates the home directory (/home/<username>) and sets the default shell to /bin/bash.
- The *useradd* command creates a user private group whenever a new user is added to the system and names the group after the user.
- **Specifying a user's full name when creating a user**: A systems administrator can use the **–c** option with *useradd* to specify the user's full name, as shown below:

```
[root@localhost bhargab]# useradd -c "Anirban Choudhury" anirban
```

Creating a user Cont'd…

- **Creating a user with the UID:** You can create a user with a custom UID with the –u option, as follows:

  [root@localhost bhargab]# useradd -u 1036 anirban

- **Creating a user with non-default home directory**: A non-default home directory can be set by executing the following command:

  [root@localhost bhargab]# useradd –d /home/test anirban

- **Adding a user to a primary group and supplementary group:** A systems administrator can specify a primary group and a supplementary one by specifying the –g and –G option, respectively.

  [root@localhost bhargab]# useradd -g "head" -G "faculty" anirban

- **Locking and unlocking a user**: A super user can lock and unlock a user account. To lock an account, one needs to invoke passwd with the -l option.

  [root@localhost bhargab]# passwd -l anirban
  Locking password for user anirban.
  passwd: Success

- The –u option with passwd unlock an account, as shown below:

  [root@localhost bhargab]# passwd -u anirban
  Unlocking password for user anirban.
  passwd: Success

Creating a user Cont'd…

- **Changing a user name:** The –*l* option with the *usermod* command changes the login (user) name, as shown below:

  [root@localhost bhargab]# usermod -l "nishant" anirban

- **Removing a user**: Combining *userdel* with the –*r* option drop a user and the home directory associated with that user, as shown below:

  [root@localhost bhargab]# userdel -r nishant

Linux Groups

- Linux group is a mechanism to organize a collection of users.
- Like the user ID, each group is also associated with a unique ID called the GID (group ID).
- There are two types of groups – a **primary** and a **supplementary** group.
- Each user is a member of a primary group and of zero or ‘more than zero’ supplementary groups.
- The group information is stored in **/etc/group** and the respective passwords are stored in the /**etc/gshadow** file.
- Some operations such as creating, deleting and modifying on a group are discussed next.

Linux Groups Cont'd…

- **Creating a group with default settings**: To add a new group with default settings, run the *groupadd* command as a root user, as shown below:

  [root@localhost bhargab]# groupadd employee

- If you wish to add a password, then type *gpasswd* with the group name, as follow:

  [root@localhost bhargab]# gpasswd employee
  Changing the password for group employee
  New Password:
  Re-enter new password:

- **Creating a group with a specified GID**: To explicitly specify the GID of a group, execute the *groupadd* command with the *–g* option, as follow:

  [root@localhost bhargab]# groupadd -g 1200 manager

- **Removing group password**: To remove a group password, run *gpasswd –r* with the relevant group name, as follow:

  [root@localhost bhargab]# gpasswd -r employee

Linux Groups Cont'd…

- **Changing the group's name:** To change the group's name, run the *groupmod* command with the *-n* option as a super user, as shown below:

  [root@localhost bhargab]# groupmod -n hrmanager employee

- **Changing the group's GID**: To change the GID of a group, run the *groupmod* command with *–g*, as follow:

  [root@localhost bhargab]# groupmod -g 1050 manager

- **Deleting a group**:
- Before deleting a primary group, delete the users of that primary group.
- To delete a group, run the groupdel command with the group name, as shown below:

  [root@localhost bhargab]# groupdel employee

Assignment

1. List two users created during the installation of Linux.
2. Give the commands which you will use to:
   a) Create and set the password of a new user
   b) Make you change to the root user.
3. If you want any command to be done with administrative or root privileges, you can use the ___ command.